

Uncertainty Quantification — Concepts

The problem

Language models produce fluent, confident-sounding text whether or not they know the answer. Uncertainty quantification asks a narrower, answerable question: can we compute a number, at generation time, that ranks wrong answers above right ones?

Two granularities

TOKEN-LEVEL one greedy decode, per-step next-token distribution p_t
entropy $H(p_t) = -\sum_v p_t(v) \log p_t(v)$ [nats]
max-softmax $\max_v p_t(v)$ margin $\text{top1} - \text{top2}$
log-perplexity $-(1/T) \sum_t \log p_t(y_t)$

SENTENCE-LEVEL N stochastic samples, compare their MEANINGS
semantic entropy H over meaning-cluster probabilities
lexical entropy H over exact-match strings (the ablation)
self-consistency share of samples equal to the modal answer

Semantic entropy — the key idea

Sampling the same question ten times may give ten different strings that all mean the same thing. Counting strings overstates uncertainty. Kuhn et al. (2023) cluster samples by bidirectional NLI entailment: answers a and b share a meaning class when a entails b AND b entails a . Entropy is then taken over cluster probabilities, not string frequencies.

Discrete $SE = -\sum_c (n_c/N) \log(n_c/N)$ cluster counts
Weighted $SE = -\sum_c P(c) \log P(c)$, $P(c) \propto \sum_{i \in c} \exp(\log p_i)$

Two traps: premise must include the QUESTION
 read the entailment index from `config.id2label`
 (cross-encoders order [contradiction, entailment, neutral])

Evaluation metrics

AUROC $P(\text{uncertainty of a wrong answer} > \text{that of a right one})$.
Rank-based (Mann-Whitney U), ties get average ranks.
 0.5 = useless. Undefined if all-correct or all-wrong.

ECE $\sum_b (n_b/N) |acc(b) - conf(b)|$. Needs a PROBABILITY, so
raw entropies are mapped through a dev-fitted calibrator.
Fixed-width bins go empty at small N -> also use equal-mass.

AURC mean risk over all coverage levels, sorting by confidence.
Lower is better. Directly models answer-or-abstain.

The leakage rule

Thresholds, calibrators and operating points are fitted on the dev split and reported on the held-out test split. Fitting on the split you report makes every method look better calibrated and every abstention threshold look more transferable than it is.

This Run — nq_open

Signal ranking

greedy accuracy 0.140 dev 100 / test 100

signal	cost	AUROC	95% CI

Sequence log-prob	single_pass	0.945	[0.899, 0.981]
Sequence probability	single_pass	0.945	[0.899, 0.981]
Lexical entropy	multi_sample	0.916	[0.848, 0.968]
Self-consistency	multi_sample	0.914	[0.842, 0.966]
Distinct answer strings	multi_sample	0.914	[0.846, 0.965]
Log-perplexity	single_pass	0.914	[0.855, 0.964]
Min max-softmax	single_pass	0.914	[0.849, 0.962]
Distinct meanings	multi_sample	0.907	[0.841, 0.958]

Selective answering

dev-fitted thresholds applied to held-out data:

target 25% -> coverage 10.0%, accuracy 0.600

target 40% -> coverage 7.0%, accuracy 0.714

target 50% -> abstained on everything

target 60% -> abstained on everything

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Signal ranking

greedy accuracy 0.560 dev 50 / test 50

signal	cost	AUROC	95% CI

Lexical entropy	multi_sample	0.884	[0.780, 0.966]
Self-consistency	multi_sample	0.884	[0.789, 0.965]
Distinct answer strings	multi_sample	0.874	[0.770, 0.961]
Sequence log-prob	single_pass	0.828	[0.684, 0.945]
Sequence probability	single_pass	0.828	[0.684, 0.945]
Semantic entropy (likelihood-weighted)	multi_sample	0.806	[0.674, 0.926]
Semantic self-consistency	multi_sample	0.800	[0.661, 0.918]
Semantic entropy	multi_sample	0.794	[0.659, 0.911]

Selective answering

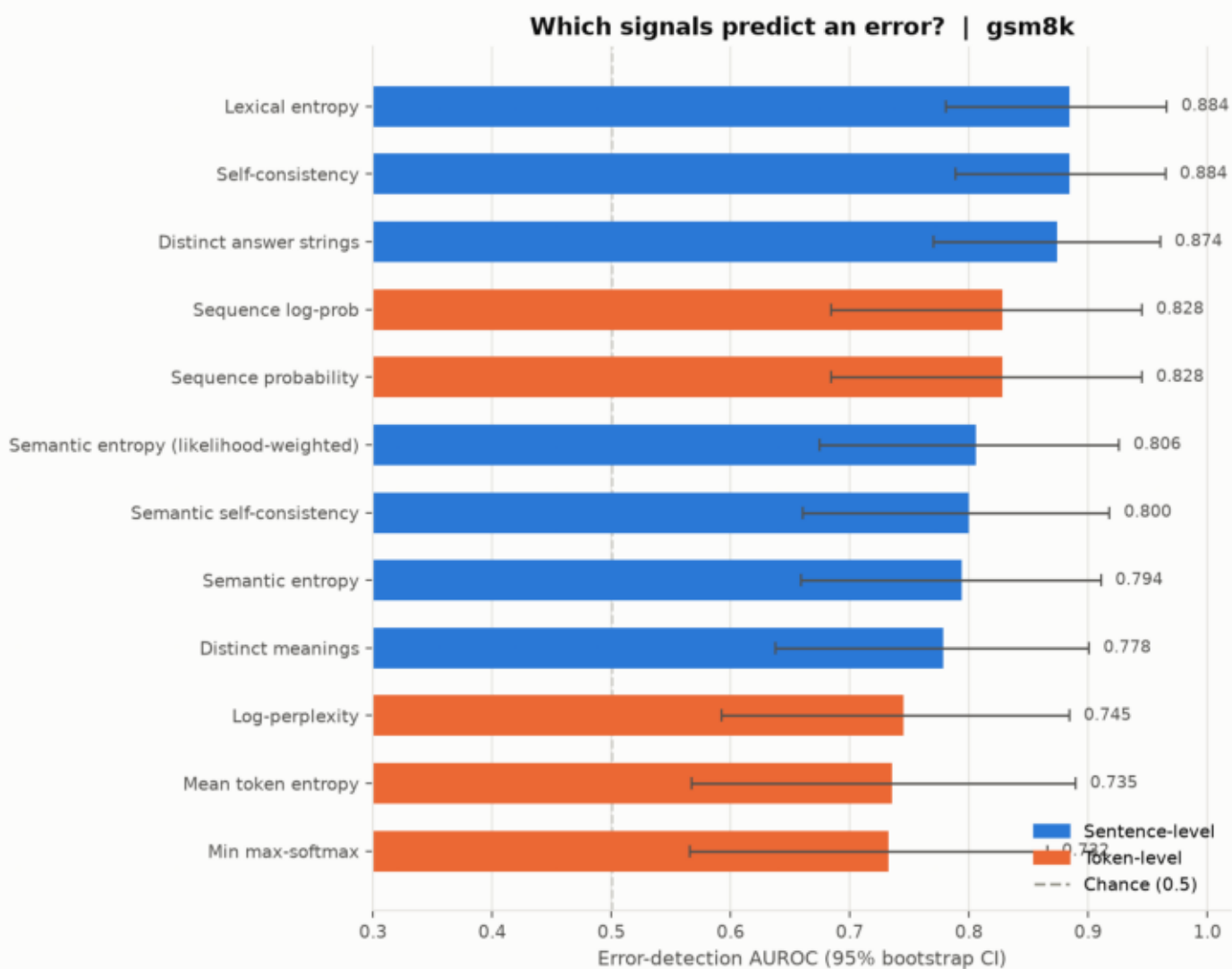
dev-fitted thresholds applied to held-out data:

target 60% -> coverage 100.0%, accuracy 0.560
target 70% -> coverage 74.0%, accuracy 0.730
target 80% -> coverage 70.0%, accuracy 0.771
target 90% -> coverage 44.0%, accuracy 0.864

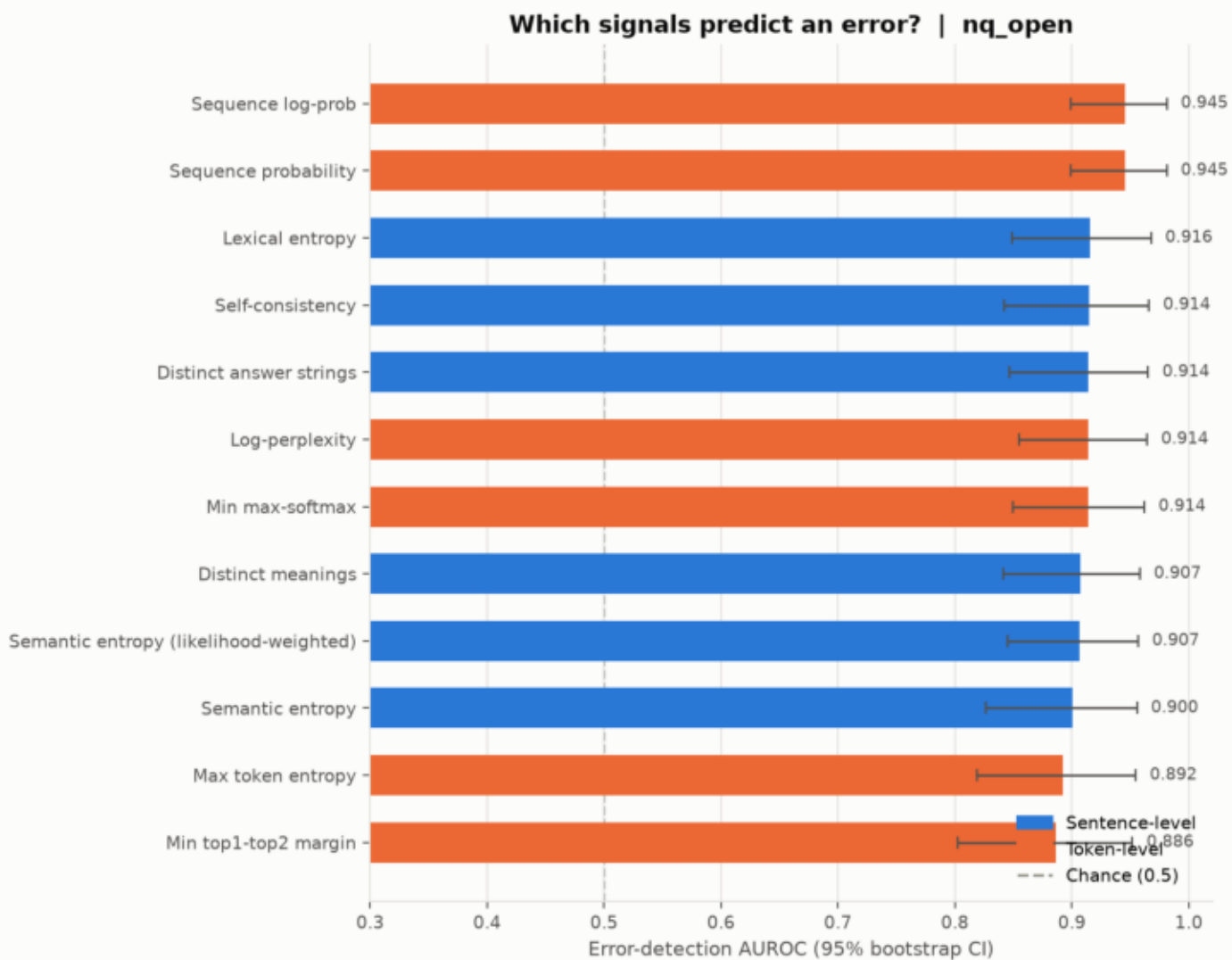
Self-consistency cascade

greedy	0.560	1.00 gen	
always sample	0.660	11.00 gen	
gated cascade	0.640	7.80 gen	(68% escalated)

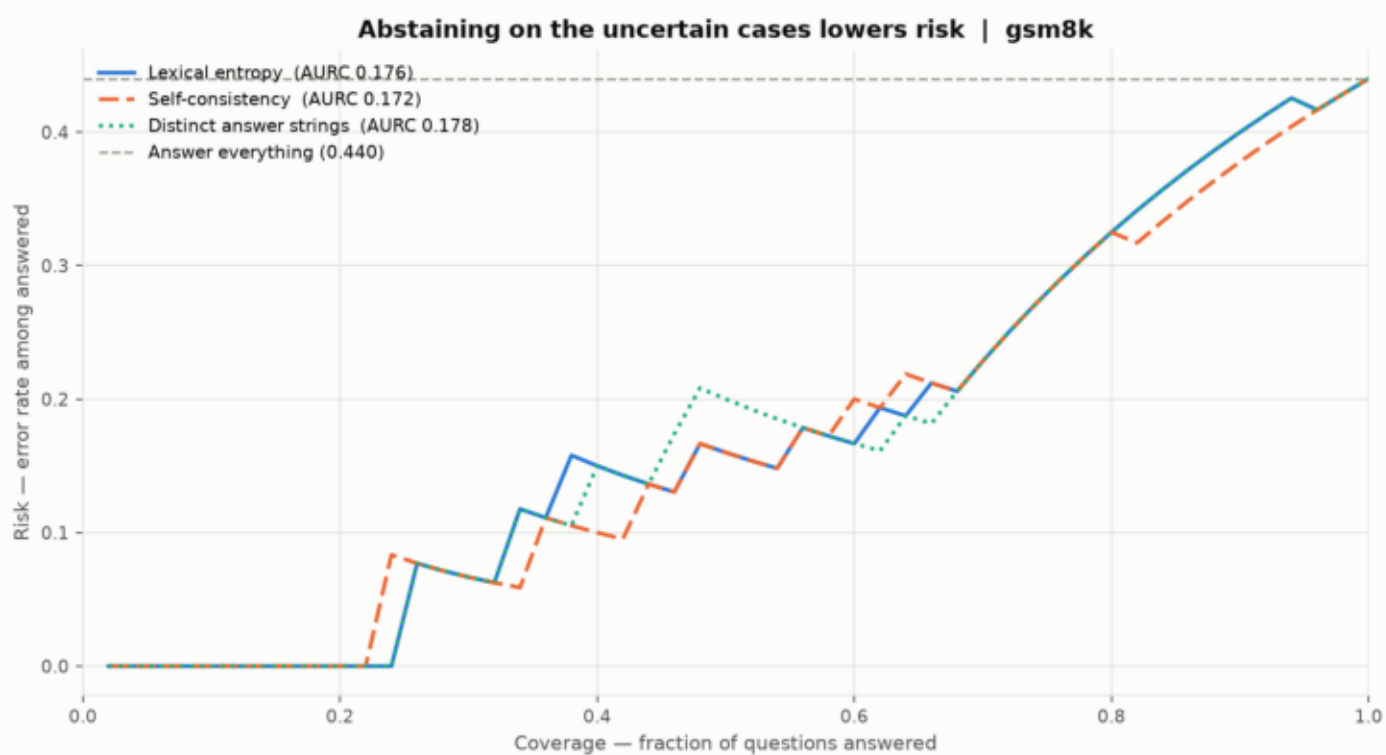
Which signals predict an error? — gsm8k



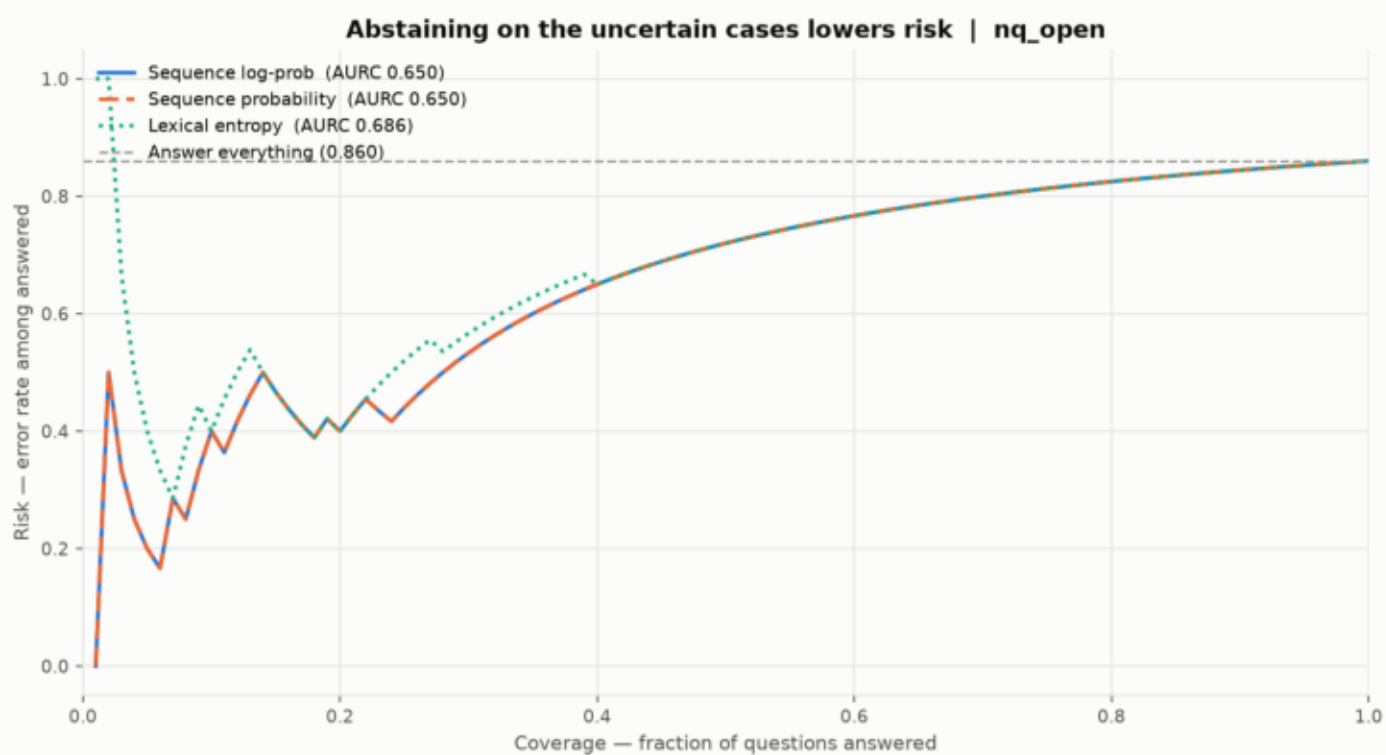
Which signals predict an error? — nq_open



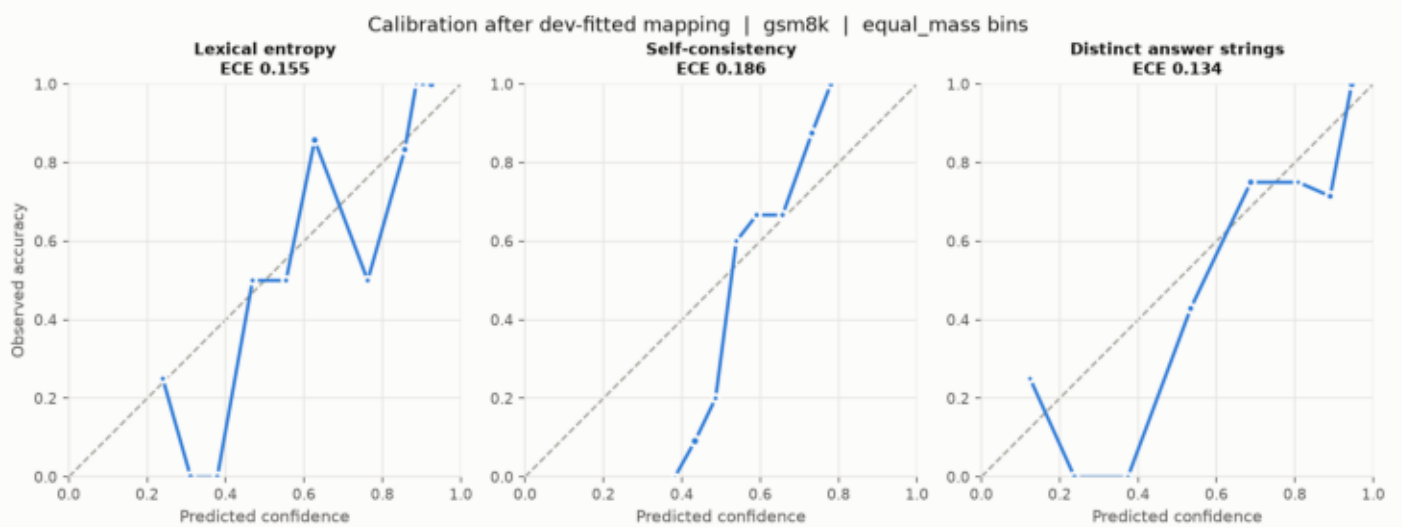
Risk falls as you abstain on the uncertain cases — gsm8k



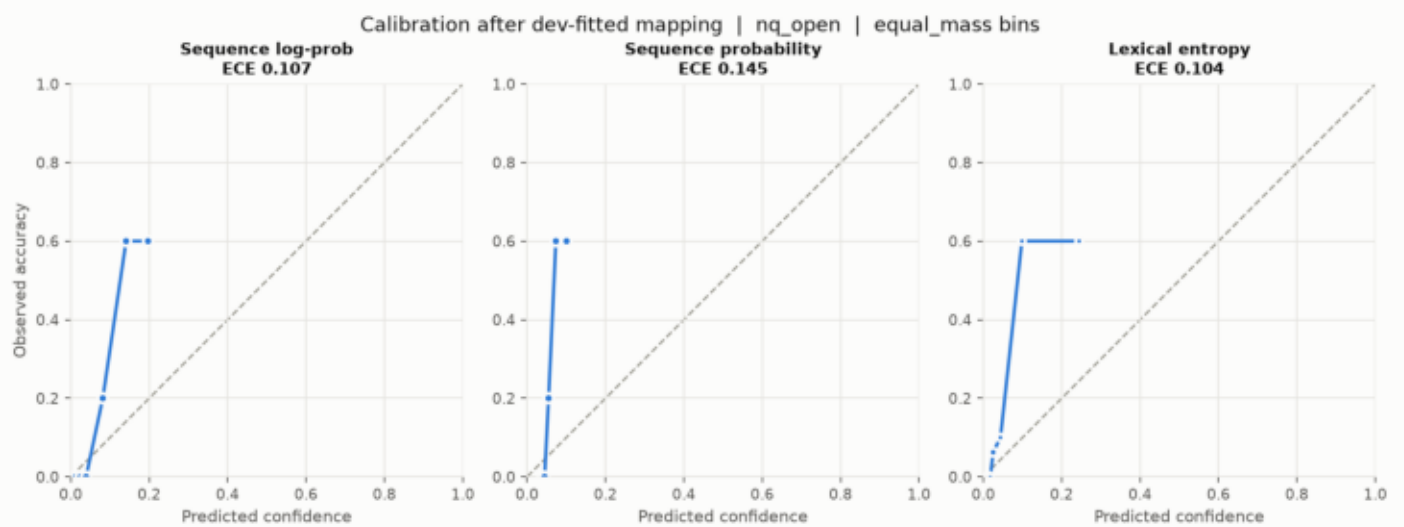
Risk falls as you abstain on the uncertain cases — nq_open



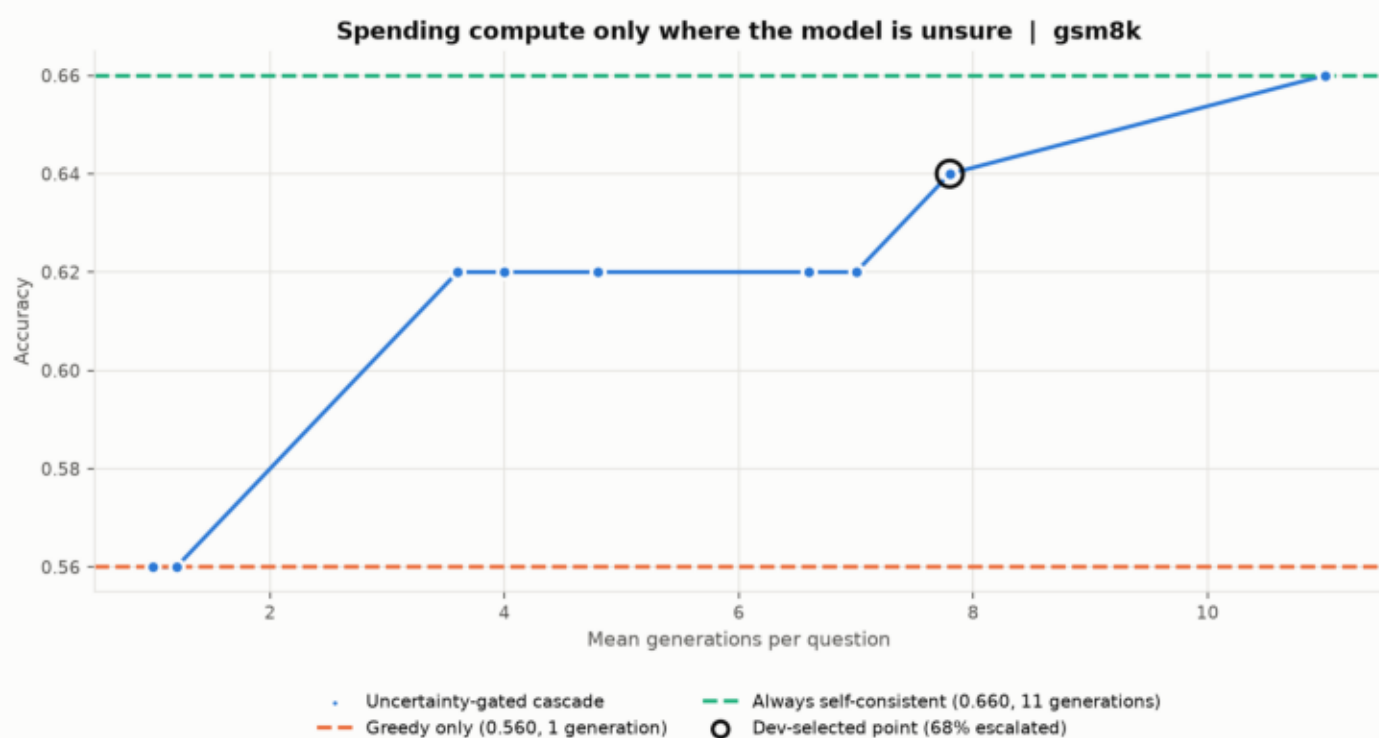
Calibration, equal-mass bins — gsm8k



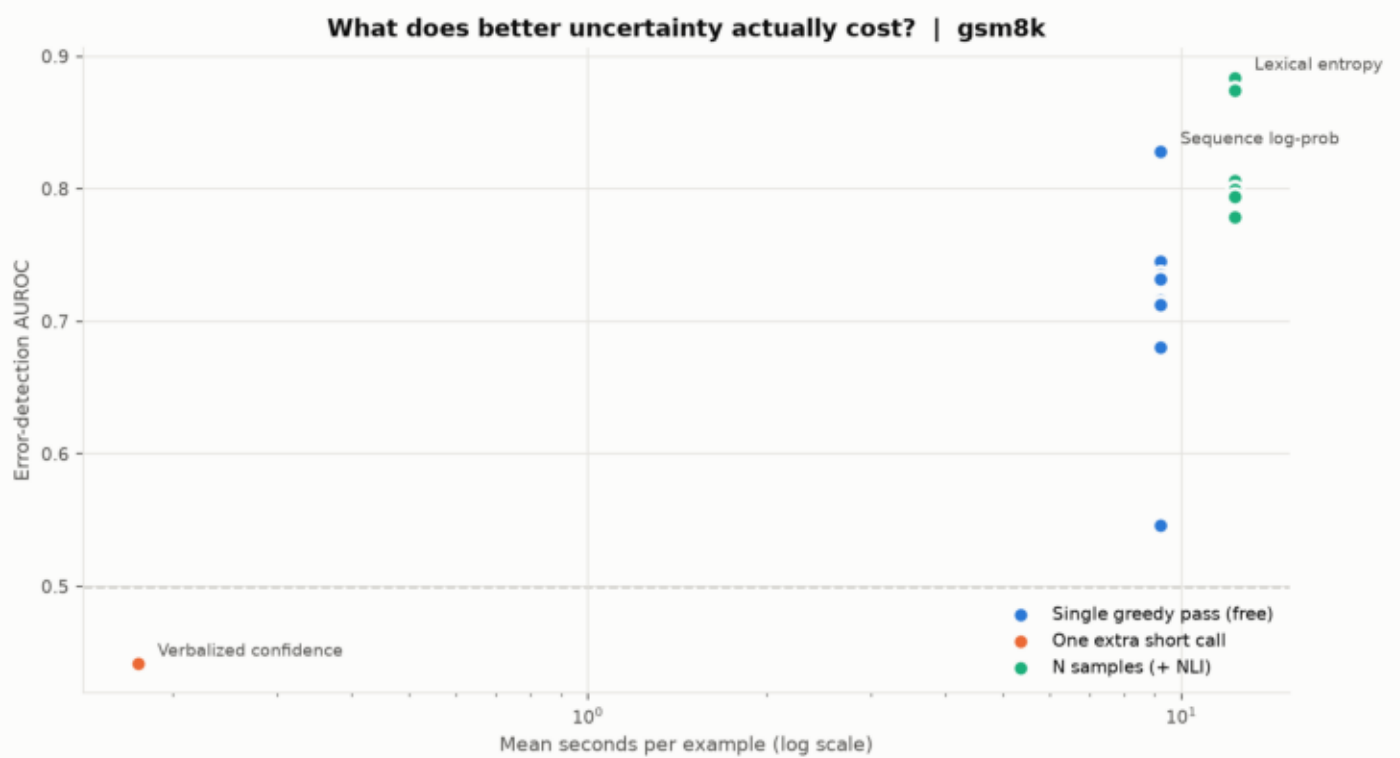
Calibration, equal-mass bins — nq_open



Spending compute only where the model is unsure — gsm8k



What better uncertainty actually costs — gsm8k



What better uncertainty actually costs — nq_open

